WW-V-51F
March 4, 1977
SUPERSEDING
Fed. Spec. WW-V-51E
November 30, 1973

FEDERAL SPECIFICATION

VALVE, ANGLE, CHECK, AND GLOBE, BRONZE (125, 150 AND 200 POUND) THREADED END, FLANGE ENDS, SOLDER ENDS, AND BRAZED END, FOR LAND USE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers copper alloy globe, angle and check valves.

1.2 Classification.

1.2.1 Types, classes, and end connections. Valves furnished under this specification shall be of the following types, classes, and end connections, as specified (see 6.2):

Type I - Globe.
II - Angle.
III - Lift check (horizontal).
IV - Swing check.

Class A - 125 pound (125 p.s.i., 8.6 Bars)
B - 150 pound (150 p.s.i., 10.3 Bars)
C - 200 pound (200 p.s.i., 13.8 Bars)

End connection 1 - Threaded, classes A, B, and C.
2 - Flanged, classes B and C.
3 - Soldered, classes A and B.
4 - Brazed, classes A, B, and C.

1.2.2 Nominal pipe sizes. Size of valves shall be as follows, and as specified (see 6.2).

Threaded, solder, and brazed end valves.

1/8, 1/4, 3/8, 1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, and 3 inches.

Flanged valves.

1, 1-1/4, 1-1/2, 2, 2-1/2, and 3 inches.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

WW-F-406 - Flanges, Cast-Iron (Classes 125 and 250) and Bronze (Classes 150 and 300).

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

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(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-V-3 - Valves, Fittings and Flanges Except for Systems Indicated Herein, Packaging of.
MIL-F-1183 - Fitting, Tube, Cast Bronze, Silver Brazing.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American National Standards Institute, Inc. (ANSI) Standards:

B2.1 - Pipe Threads (Except Dryseal).
B16.18 - Cast Bronze Solder-Joint Pressure Fittings.
B16.24 - Bronze Flanges and Flanged Fittings.

(Application for copies should be addressed to the American Standards Institute, Inc. 1430 Broadway, New York, NY 10018.)

American Society for Testing and Materials (ASTM) Standards:

A276 - Stainless and Heat-Resisting Steel Bars and Shapes.
A296 - Corrosion-Resistant Iron-Chromium, Iron-Chromium-Nickel, and Nickel Base Alloy Coatings for General Applications.
B16 - Free-Cutting Brass Rod, Bar, and Shapes for Use on Screw Machines.
B61 - Steam or Valve Bronze Castings.
B62 - Composition Bronze or Ounce Metal Castings.
B98 - Copper-Silicon Alloy Rod, Bar and Shapes.
B99 - Copper-Silicon Alloy Wire for General Purposes.
B124 - Copper and Copper Alloy Forgings, Rod, Bar, and Shapes.
B140 - Leaded Red Brass (Hardware Bronze, Rod, Bar, and Shapes.)
B148 - Aluminum Bronze Sand Castings.
B149 - Leaded Nickel Brass and Leaded Nickel Bronze Sand Castings.
B150 - Aluminum Bronze Rod, Bar, and Shapes.
B164 - Nickel-Copper Alloy Rod and Bar.
B371 - Copper-Zinc-Silicon Alloy Rod.
B505 - Copper-Base Alloy Continuous Casting.
B584 - Copper Alloy Sand Castings for General Applications.
D2240 - Indentation Hardness of Rubber and Plastics by Means of a Durometer.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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Manufacturers Standardization Society of the Valve and Fittings Industry:

MSS-SP-25 - Standard Marking System for Valves, Flanges, Fittings and Unions.

(Application for copies should be addressed to the Manufacturers Standardization Society of the Valve and Fittings Industry, 1815 North Fort Myer Drive, Arlington, VA 22209).

3. REQUIREMENTS

3.1 Material.

3.1.1 Valve body, all types.

3.1.1.1 Classes A and B, with metallic, and nonmetallic disc, and class C with nonmetallic disc. The valve body shall be copper alloy 836 or 922 meeting the chemical requirements of ASTM B62 or B61 respectively (see 4.3.1).

3.1.1.2 Class C with metallic disc. The valve body shall be copper alloy 922, meeting the chemical requirements of ASTM B61 (see 4.3.1).

3.1.2 Types I and II, bonnet, and types III and IV, cap.

3.1.2.1 Classes A and B, with metallic, and nonmetallic disc, and class C with nonmetallic disc. Bonnets or caps shall be any of the following copper alloys and shall meet the chemical requirements of the following ASTM designations for all valve sizes (see 4.3.1):

COPPER ALLOY	ASTM
377	B124
675	B124
836	B62 or B584
922	B61 or B584

3.1.2.2 Class C with metallic disc. Bonnets or caps shall be copper alloy 922 meeting the chemical requirements of ASTM B61 for all valve sizes (see 4.3.1).

3.1.2.3 Classes A, B, and C, with metallic, and nonmetallic disc. Copper alloy 360 meeting the chemical requirements of ASTM B16 may be used for the bonnet or cap on valve sizes 3/4 and smaller in size (see 4.3.1).

3.1.3 Types I, II and III, union ring or nut and type IV, screw-in cap.

3.1.3.1 Classes A and B, with metallic, and nonmetallic disc and class C with nonmetallic disc. Unless otherwise specified (see 6.2), the copper alloy for union ring, nut, or screw-in bonnet shall be at the manufacturer's option.

3.1.3.2 Class C with metallic disc. The union ring or nut shall be copper alloy 922 meeting the chemical requirements of ASTM B61. Copper alloy 360 meeting the chemical requirements of ASTM B16 may be used for the union ring or nut on valve sizes 3/8 and smaller in size (see 4.3.1):

3.1.4 Types I and II, stem.

3.1.4.1 Classes A, B, and C with metallic and nonmetallic disc. Stems shall be of the following alloys, and shall meet the chemical requirements of the following ASTM designations for all valve sizes (see 4.3.1):

COPPER ALLOY	ASTM	COPPER ALLOY	ASTM
642	B124 or B150	874	B584
651	B98 or B99	875	B584
694	B371	876	B584
697	B371	922	B61
836	B62	956	B148
864	B62		

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3.1.4.2 Classes A, B and C, valve sizes 1/2 and smaller. Unless otherwise specified (see 6.2), classes A, B, and C, shall meet the chemical requirements of ASTM B61 or alloy 922 of ASTM B505 or alloys specified in 3.1.4.1 (see 4.3.1):

3.1.5 Types I, II, III and IV, discs.

3.1.5.1 Class A, metallic discs. Discs shall be any of the following copper alloys, and shall meet the chemical requirements of the following ASTM designations (see 4.5.1):

COPPER ALLOY	ASTM	COPPER ALLOY	ASTM
316	B140	922	B61
320	B140	973	B149
642	B124 or B150	976	B149
675	B124	978	B149
936	B62		

Copper alloy 360 meeting the chemical requirements of ASTM B16 may be used for discs in valves size 3/4 and smaller.

3.1.5.2 Class B, discs. Unless otherwise specified (see 6.2), discs for class B valves shall be nonmetallic.

3.1.5.3 Class C, metallic discs. Discs shall be copper alloy, nickel-copper alloy, nickel-copper-silicon alloy, or stainless steel, as specified (see 4.3.1 and 6.2):

(a) Copper alloy discs. Copper alloys shall meet the chemical requirements of the following ASTM designations:

COPPER ALLOY	ASTM	COPPER ALLOY	ASTM
314	B140	922	B61
642	B124 or B150	973	B149
651	B98 or B99	976	B149
836	B62	978	B149

(b) Nickel-copper alloy. Wrought discs shall meet the chemical requirements for class A of ASTM B164. Cast discs shall meet the chemical requirements for grade M-35 or M35W of ASTM A296.

(c) Nickel-copper-silicon discs. Discs shall contain 60 percent or more of nickel, and silicon in amounts of 2.7 to 3.7, 3.3 to 4.3, or 3.5 to 4.5 percent, as specified (see 6.2).

(d) Stainless steel meeting the requirements of ASTM A276, type 416, may be used for discs in valves size 3/4 and smaller.

3.1.5.4 Classes A, B and C, nonmetallic disc. Unless otherwise specified (see 6.2), material for nonmetallic discs shall be at the manufacturer's option except that material shall meet the requirements of 3.2.3.

3.1.6 Types I, II, III and IV, disc holder or guide.

3.1.6.1 Class A. The disc holder shall be copper alloy. Unless otherwise specified (see 6.2), the type of copper alloy shall be at the manufacturer's option.

3.1.6.2 Classes B and C. Disc holders shall be any of the following copper alloys, and shall meet the chemical requirements of the following ASTM designations (see 4.3.1):

COPPER ALLOY	ASTM	COPPER ALLOY	ASTM
316	B140	836	B62
320	B140	922	B61
377	B124		

Copper alloy 360 meeting the chemical requirements of ASTM B16 may be used for the disc holders in valves size 3/4 and smaller.

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3.1.7 Types I, II and III, seats. Unless otherwise specified valve seats shall be integral or inserted at the manufacturers option (see 6.2).

3.1.8 Types I and II, packing nut and follower. Packing nuts and followers shall be commercial copper alloy of suitable composition.

3.1.9 Types I and II, handwheels.

3.1.9.1 Class A valves. Handwheels shall be gray, malleable or ductile iron, steel, aluminum alloy die casting, or zinc alloy die casting.

3.1.9.2 Class B and C valves. Handwheels shall be gray, malleable or ductile iron, steel or aluminum alloy.

3.1.9.3 Class A, B and C, large handwheels. Handwheels 6 inches (152 mm) in diameter and larger shall be gray, malleable or ductile iron, steel, or cast aluminum alloy.

3.2 Construction and design. All valves shall be of a construction for resisting distortion as well as pressure under normal service conditions. The design shall provide rounded corners and surfaces on valve bodies and bonnets, except at ends. When valve is wide open, the area of the passage through the valve at all points shall be not less than the area of a circle having a diameter equal to the nominal pipe size of the valve, except that the area through seat opening may be reduced by the guides for the disc or driving lugs for seats.

3.2.1 Body and bonnet. Bonnet joint shall be inside-screw, one-piece screw-over, or union-ring construction with contact surfaces of bonnet and body machined so as to provide a pressure-tight metal to metal joint without a gasket, except that the bonnet joint for valves sizes 2-1/2 and larger may be flanged and provided with a suitable gasket. Types I and II valve bonnets shall have a machined surface located below the stuffing box to contact the disc stem ring or back seating surface on the stem when the valve is wide open, to permit repacking while under pressure (see 4.3.3).

3.2.2 Seat and disc.

3.2.2.1 Types I and II, globe and angle valves.

3.2.2.1.1 Class A valves. Valves shall have the disc or disc holder disc securely fastened to the end of the stem with a stem ring which allows the disc or disc holder to swivel freely, except that for valves size 1/2 or smaller the disc may be formed on the end of the stem.

3.2.2.1.2 Class B valves. Valves shall have a slotted disc holder to slip on over an enlarged section of the disc end of the stem, except that valves size 2-1/2 and larger may be furnished with a disc stem ring connection between the stem and disc holder. The construction shall allow the disc holder to swivel freely. Disc holders of the slip-on type shall be guided for self-centering and shall be so constructed with relation to the body and bonnet that when the valve is wide open, the disc holder does not slip off the stem during assembling and disassembling. For valve sizes 1/2 and smaller, the disc may be formed on the stem. Unless otherwise specified (see 6.2), valve seats shall be integral or inserted at the manufacturer's option.

3.2.2.1.3 Class C valves. Valves with metallic discs shall have a renewable alloy seat ring with a swivel alloy disc of the locknut type, except that for valve sizes 1/2 and smaller, the disc may be formed on the end of the stem. Valves with nonmetallic discs may have an integral seat and shall have a disc holder of the slip-on or locknut type. The slip-on type shall be guided for self-centering on the seat and constructed in such a manner that they do not slip off the stem when the valve is in the wide open position during assembly and disassembly.

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3.2.2.2 Classes A, B and C valves.

3.2.2.2.1 Type III, lift check (horizontal). Valves shall have the disc or disc holder guided to insure proper disc seating. Seats for classes A and B may be integral or renewable. Discs for classes A and B shall be metallic or nonmetallic at the manufacturer's option. Class C valves shall have renewable seat and an alloy metal disc, unless otherwise specified (see 6.2).

3.2.2.2.2 Type IV, swing check. Valve shall have a disc, or disc holder securely fastened to the hinge. The connection shall allow sufficient play so that the disc properly seats by gravity and swivels to distribute wear uniformly, except that the disc and hinge for class A valves may be in one piece. When in the wide open position, the disc or disc holder to hinge assembly shall bear on a stop. When specified (see 6.2), class A valves shall have a leather or other type of nonmetallic material disc facing for cold water.

3.2.3 Nonmetallic discs. Discs shall be suitable for the service specified (see 6.2), and shall conform to the applicable requirements for hardness given in table I, (see 4.5.2.3.1).

TABLE I. Hardness, nonmetallic discs

Indentation	Service and designation	
	Hot and cold water HW and CW	Air and gas A and G
Minimum indentation, mm	0.01	0.10
Maximum indentation, mm	.65	1.05

3.2.4 Types I and II, stem. Each valve stem shall be threaded with either an Acme form thread or a 60° thread approximating the stub thread form so that the valve is open when handwheel or stem is rotated counterclockwise. The length of thread engagement of the stem shall be not less than the outside diameter of the thread when the valve is closed, and not less than 75 percent of the outside diameter of the thread when the valve is open. (Valves shall be provided with a backseating surface on bonnet to seal against stem or disc stem ring when the valve is in the wide open position. Unless otherwise specified (see 6.2), stem diameters shall be in accordance with table II. The handwheel end of the valve stem shall be square and tapered with its greatest diagonal approximately equal to the diameter of the stem or it may be of splined or serrated design. A threaded extension shall be provided to accommodate a wheel nut.

TABLE II. Minimum stem diameters, globe and angle valves

Nominal pipe size	Diameter of stem 1/					
	Class A		Class B		Class C	
	Inch	mm	Inch	mm	Inch	mm
1/8	0.23	5.3	0.24	6.1	0.28	7.1
1/4	.24	6.1	.28	7.1	.28	7.1
3/8	.28	7.1	.28	7.1	.28	7.1
1/2	.31	7.9	.35	8.9	.35	8.9
3/4	.35	8.9	.39	9.9	.40	10.2
1	.39	9.9	.42	10.7	.44	11.2
1-1/4	.42	10.7	.49	12.4	.50	12.7
1-1/2	.49	12.4	.53	13.5	.53	13.5
2	.53	13.5	.59	15.0	.62	15.7
2-1/2	.59	15.0	.67	17.0	.67	17.0
3	.67	17.0	.75	19.0	.75	19.0

1/ Minimum diameter of stem refers to the diameter of that part of the stem which passes through the stuffing box (see 3.2.5.1).

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3.2.5 Types I and II, stuffing box and packing.

3.2.5.1 Stuffing box. The stuffing box may be either the gland-follower type or the packing-nut type on all class A valves, and on classes B and C valves size 1/2 and smaller. Classes B and C valves size 3/4 and larger shall have a stuffing box of the gland follower type. The length of packing in contact with stem shall be at least one diameter of the stem passing through it. The thread shall be sufficient length to enable the nut to be screwed on the stuffing box at least two full turns when the full amount of packing is in place. The thread length shall be sufficient for full travel of the follower. Valves shall be constructed so that packing can be added to the stuffing box or repacked when valve is wide open and under pressure (see 4.3.3).

3.2.5.2 Packing. Unless otherwise specified (see 6.2), valves shall be packed for water service, with graphite impregnated asbestos or tetrafloroethylene-impregnated asbestos valve stem packing of braided, twisted, or formed ring type.

3.2.6 Types I and II, handwheels. Handwheels smaller than 6 inches (152 mm) in diameter shall be the ventilating (nonheat) design with raised rim to protect the hand, and shall have three or more spokes. Handwheels 6 inches (152 mm) and larger may be the plain rim type marked to show the opening direction. Unless otherwise specified (see 6.2), diameter of handwheels shall be in accordance with table III.

TABLE III. Minimum diameter of handwheels, globe and angle valves

Nominal pipe size	Diameter of handwheels 1/					
	Class A		Class B		Class C	
	Inches	mm	Inches	mm	Inches	mm
1/8	1.50	38	1.50	38	1.50	38
1/4	1.50	38	1.75	44	1.75	44
3/8	1.75	44	2.00	51	2.00	51
1/2	2.00	51	2.25	57	2.50	64
3/4	2.50	64	2.62	66	2.75	70
1	2.75	70	3.00	76	3.00	76
1-1/4	3.00	76	3.50	89	3.50	89
1-1/2	3.50	90	3.75	95	4.00	102
2	4.00	102	4.38	111	4.75	120
2-1/2	4.75	120	5.00	127	5.38	137
3	5.38	137	5.75	146	6.00	152

1/ Handwheel diameters shall not exceed the specified minimum diameter by more than 50 percent. When irregular handwheels are furnished, the largest diameter of the shape shall be considered the handwheel diameter.

3.2.7 Types I, II, III and IV, valve-end connection.

3.2.7.1 Connection 1, threaded. The ends shall be threaded internally in accordance with ANSI B2.1. Threads shall be clean, smooth, true to form and concentric with the axis of the valves. Variations in alignment of threads shall not exceed 1/16 inch per foot. Threads shall be chamfered a distance not less than one-half the pitch of the threads at an angle of approximately 45° with the axis of the thread for the purpose of easy entrance in making a joint and for the protection of the thread. The chamfer shall be concentric with the thread, and shall be included in measurements of depth of thread chamber and thread length. All threads shall conform to table IV.

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TABLE IV. Connection 1, thread dimensions

Nominal pipe size	Length of thread 1/		Depth of thread chamber, minimum 2/	
	Inch	mm	Inches	mm
1/8	0.25	6.4	0.35	8.9
1/4	.28	7.1	.40	10.2
3/8	.31	7.9	.42	10.7
1/2	.37	9.4	.54	13.7
3/4	.44	11.2	.57	14.5
1	.51	13.0	.70	17.8
1-1/4	.60	15.2	.71	18.0
1-1/2	.62	15.8	.75	19.0
2	.67	17.0	.79	20.1
2-1/2	.82	20.8	1.14	29.0
3	.88	22.4	1.20	30.5

1/ Distance parallel to axis from face of valve to inner end of thread.
2/ Distance parallel to axis from face of valve to nearest point of seat diaphragm wall.

3.2.7.2 Connection 2, flanged. Flanges shall conform to WW-F-406 or ANSI B16.24. Unless otherwise specified (see 6.2), flanged shall be faced and drilled.

3.2.7.3 Connection 3, solder. Solder-end valve connections shall conform to ANSI B16.18.

3.2.7.4 Connection 4, brazed welding ends. Dimensions of brazed-end valve connections shall conform to type B in table III of MIL-F-1183.

3.3 Types I, II, III and IV.

3.3.1 Temperature ratings. Unless otherwise specified (see 6.2), temperature ratings for valves shall be at the manufacturer's option.

3.3.2 Finish. Unless otherwise specified (see 6.2), valves shall have a natural or machined finish produced by commercial manufacturing process.

3.4 Performance requirements.

3.4.1 Hydrostatic test.

3.4.1.1 Types I and II. The shell shall not deform or leak, and the leak rate through the valve seat shall not exceed 10 ml. per hour per inch of valve size for valves size 1 or larger. Valves smaller than size 1 shall not have a leak rate through the valve seat to exceed 10 ml per hour (see 4.3.2.1 and 4.3.2.4).

3.4.1.2 Types III and IV. The shell shall not deform or leak, and the leak rate through the valve seat shall not exceed 40 ml. per hour per inch of valve size for valves size 1 or larger. Valves smaller than size 1 shall not have a leak rate through the valve seat to exceed 40 ml. per hour (see 4.3.4).

3.4.2 Pneumatic test.

3.4.2.1 Types I and II. The shell shall not leak, and the leak rate through the valve seat shall not exceed 0.10 standard cubic feet (2.83 liters) of air per hour per valve size (see 4.3.2).

3.4.2.2 Type III and IV. The shell shall not leak, and the leak rate through the valve seat shall not exceed 0.40 standard cubic feet (11.33 liters) of air per hour per inch of pipe size (see 4.3.4).

3.4.3 Steam tests. The shell shall not leak and there shall be no visible leakage from the seat (see 4.3.2.1 and 4.3.2.2).

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3.4.4 Types I, II, III and IV, nonmetallic discs.

3.4.4.1 Hardness. Discs shall not crack, chip, distort, or be otherwise adversely affected. The disc shall seat when the valve is closed and shall maintain a seal and shall not leak under pressure (see 4.3.5.1).

3.4.4.2 Dry heat. Discs shall not disintegrate or blister when tested in accordance with 4.3.5.2.

3.4.5 Types I and II, stuffing box packing. When tested in accordance with 4.3.3. Leakage shall not exceed 240 ml. per minute on removal of the packing. There shall be no leakage after repacking as specified.

3.4.6 Steam test. Nonmetallic discs shall not disintegrate or blister when tested in accordance with 4.3.5.3.

3.5 Marking. All valves shall be legibly marked in accordance with the requirements of MSS SP-25.

3.6 Workmanship. All parts of valves shall be new and free from injurious defects, and shall be fitted and finished in a workmanlike manner. Castings shall be free from blowholes, in line surface porosity, hard spots, cracks, or other injurious defects. They shall be smooth and cleaned, both inside and outside, and parting line and similar roughness shall be removed. Castings shall not be repaired, plugged, brazed or "Burned in". Finish shall be free from noticeable imperfections.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Sampling.

4.2.1.1 Sampling for visual and dimensional examination. Sampling for these characteristics shall be in accordance with the provisions of MIL-STD-105 with the inspection levels and AQL's as specified in 4.2.2.1.

4.2.1.2 Sampling for tests. Sampling for tests shall be selected in accordance with the provisions of MIL-STD-105 inspection level S-1. The acceptable quality (AQL) shall be 2.5 percent defective.

4.2.2 Examination.

4.2.2.1 Inspection levels and acceptable quality levels (AQL's). The inspection levels and AQL's expressed in defects per 100 units shall be as follows:

		AQL	
	Level	Major	Minor
For visual examination.	II	1.5	4.0
For dimensional examination.	S-2	-	4.0

4.2.2.2 Visual examination. Samples selected in accordance with 4.2.1 shall be visually examined as specified in table V and examined for dimensions as specified in tables II, III and IV, for dimensions in WW-F-406 for flanged valves, and in ANSI B16.18 for solder-end valves. Any dimension not within specified tolerances shall be classified as a defect.

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TABLE V. Classification of defects

Examine	Defects	Major	Minor
Design	Not type, class, end connection and size specified	X	
Construction and workmanship (as applicable)	Component parts missing	X	
	Component cracked, fractured, deformed, seamed, or with laminations, metal slivers, sharp edges or burrs, tears, holes or other injurious defects (exterior and interior surfaces)	X	
	Cast component not sound and smooth, porosity, fins, cracks, blowholes, depressions, hard spots, bumps or lumps, mold marks of parting line, or other injurious defects	X	
	Exterior and interior surfaces not clean, smooth (grooves, folds, indentations, flaws, grease, dirt or foreign matter imbedded in surfaces)		X
	Cast components repaired, plugged, brazed or burned in	X	
	Components not accurately and smoothly machined on contacting surfaces	X	
	Disc not securely fastened to stem with stem ring; disc does not swivel freely; disc holder of slip-on type not adequately guided for self-centering; disc holder slips off; or disc not tightly seated	X	
	Flanged valves not faced or drilled as specified	X	
	Handwheel not marked as specified		X
	Stem packing or packing in stuffing box is not as specified	X	
	Threading on screwed valves not as specified	X	
	Thread burred or damaged, or fit between threads loose	X	
Identification marking	Missing, illegible, incomplete or not in accordance with MSS-SP-25		X

4.2.3 Examination of preparation for delivery. An examination shall be made to determine that preservation, packaging, packing and marking requirements of the applicable contract or order are complied with. Defects shall be scored in accordance with table VI. The lot size shall be the number of shipping containers fully prepared for delivery with the exceptions that containers need not be sealed or closed, nor interior containers, or case liners sealed (if applicable). Examination shall be made in two phases; first as interior examination in process of packaging, and second, and examination of containers fully prepared for delivery. The sample unit for each of the two phases shall be one container prepared for delivery as set forth above. The inspection level shall be S-2, the acceptable quality level (AQL) shall be 4 percent defective.

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TABLE VI. Classification of preparation for delivery defects

Examine	Defects
Marking (interior package or exterior container as applicable)	Omitted, incorrect, illegible, improper size, location, sequence or method of application.
Materials	Components missing, damaged, defective or not as specified.
Workmanship (as applicable)	Inadequate or improper packaging or packing, such as closure of interior packages or containers, closure of case liner or container flaps, taping of seams, corners, and manufacturer's joint, closure of alternate containers; loose strapping or tape banding; inadequate stapling; bulging or distortion of containers.
Contents (interior and exterior container as applicable)	Number per container not as specified.
Weight (exterior container)	Weight per container exceeds maximum.
Preservation (as applicable)	Preservation missing, improperly applied, or incorrect type.

4.2.4 Tests. Samples selected in accordance with 4.2.1.2 shall be tested as specified in 4.3.

4.3 Test methods.

4.3.1 Chemical analysis. Analyze compositions according to test methods in the documents listed in 3.1.1.1 to 3.1.6.2.

4.3.2 Types I and II pressure tests.

4.3.2.1 Shell. Apply pressure by one of the following methods at or near room temperature (see 3.4.1.1, 3.4.2.1 and 3.4.3).

(a) Hydrostatically at twice the rated pressure (see 1.2.1).
(b) Pneumatically at 80 to 100 psig (5.6 to 6.9 bars) underwater or automatically.
(c) Steam tested at 125 psig (8.7 bars).

4.3.2.2 Seat. Apply pressure by one of the following methods at or near room temperature (see 3.4.1.1 and 3.4.2.1):

(a) Hydrostatically at the rated pressure (see 1.2.1).
(b) Pneumatically at 80 to 100 p.s.i.g (6.5 to 6.9 bars) underwater or automatically.
(c) Steam tested at 125 psig (8.7 bars).

4.3.3 Stuffing box, packing test. When specified (see 6.2), test the completely assembled valve in a fixture with the valve outlet plugged and with the valve in the wide open position. Apply hydrostatic pressure equal to the maximum working pressure through the inlet side of the valve. The water for test purpose shall not exceed 212°F(100°C). Remove the packing from the stuffing box and repack the valve maintaining the required internal hydrostatic pressure. Adjust (tighten) the gland follower or pack nut, as applicable, so no leakage occurs when the valve is being opened or closed. If unacceptable leakage occurs on removal of the packing or if the stuffing box cannot be packed under the specified conditions the valve shall be classified as defective. However, any evidence of leakage after repacking, with the valve disc open or in any other position, with the valve under the specified pressure, the valve shall be considered defective (see 3.2.1, 3.2.5.1 and 3.4.5).

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4.3.4 Types III and IV pressure tests. Apply pressure at the outlet side of the disc by one of the following methods at or near room temperature (see 3.4.2.2):

(a) Hydrostatically at 50 p.s.i.g (3.4 bars) mimimum.
(b) Pneumatically at 50 p.s.i.g (3.4 bars) minimum, underwater or automatically.

4.3.5 Types I, II, III and IV, discs, nonmetallic hardness, dry heat and steam tests. When specified (see 6.2), test for hardness, dry heat and steam test as follows.

4.3.5.1 Hardness. Test for hardness in accordance with ASTM D2240, and at a temperature of 270° ± 5°F (132° ± 2.8°C) for 60 ± 5 minutes (see 3.2.3 and 3.4.4.1).

4.3.5.2 Dry heat. Heat one half of a disc in air to a temperature of 270° ± 5°F (132° ± 2.8°C) for 60 ± 5 minutes (see 3.4.4.2).

4.3.5.3 Steam test. Expose one half of a disc to saturated steam at 400° ± 5°F for 60 ± 5 minutes (see 3.4.6).

5. PREPARATION FOR DELIVERY

5.1 Commercial packaging and packing. Unless otherwise specified (see 6.2) the valves shall be packaged in a manner to assure protection against deterioration and damage from the contractor to the initial destination. The packaged valves shall be packed to assure acceptance and safe delivery at destination in containers complying with Uniform Classification rules or National Motor Freight Classification rules.

5.2 Preservation, packaging, and packing. When specified, the valves furnished in quantities as specified (see 6.2), shall be preserved, packaged, and packed in accordance with MIL-V-3. The level of preservation and packaging shall be A or C and the level of packing shall be A, B, or C, as specified (see 6.2).

5.3 Marking. Marking shall be in accordance with 5.2.1 or 5.2.2, as specified (see 6.2).

5.3.1 Civil agencies. In addition to any special markings required by the contract or order, all interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special markings required by the contract or order, all interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Valves covered by specification are intended for normal service with liquids and gases.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, end connection, and size (see 1.2.1 and 1.2.2).
(c) Material (see 3.1.3.1, 3.1.4.2, 3.1.5.2, 3.1.5.3, 3.1.5.4 and 3.1.7).
(d) Types I and II, seat for classes A and B valves, when required (see 3.2.2.1.1 and 3.2.2.1.2).
(e) Type III, seat and disc, when required (see 3.2.2.2.1).
(f) Type IV, when and service required for class A valves (see 3.2.2.2.2).
(g) Types I to IV, designation for nonmetallic discs (see 3.2.3).
(h) Types I and II, stem diameters, and stem packing, when required (see 3.2.4 and 3.2.5.2).
(i) Types I and II, handwheel size, when required (see 3.2.6).
(j) Types I to IV, connection 2 flanges (see 3.2.7.2).
(k) Types I to IV, temperature ratings, and finish, when required (see 3.3.1 and 3.3.2).
(l) Types I and II, stuffing box test, when required (see 4.3.3).
(m) Level of preservation and packaging, and level of packing if other than specified (see 5.2).
(n) Marking (see 5.3).
(o) Nonmetallic disc tests, when required (see 4.3.5).

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6.3 Dezincification. Copper-zinc alloys (brasses) which do not contain any inhibitors such as tin, antimony, silicon, silver, phosphorus or arsenic, and which contain zinc in amounts greater than 16 percent by weight should not be used for valve parts in contact with waters having a pH less than 6 or greater than 8.

6.4 Copper alloys. Consult the following publications for information on copper alloys:

Copper Development Association (CDA) Publications:

Copper alloys 300 and 600 series.

Standard Handbook, Copper, Brass, Bronze, Wrought Mill Products, Alloy Data /2.

Standards Handbook, Copper, Brass, Bronze, Wrought Mill Products, Specification Index /6.

Copper alloys 800 and 900 series.

Standard Handbook, Copper, Brass, Bronze, Cast Products, Data Specifications /7.

(Application for copies should be addressed to the Copper Development Association, Inc. 405 Lexington Avenue, New York, NY 10017).

6.5 Solder ends, connection 3. The maximum recommended pressure temperature ratings for solder end connections should conform to the requirements of ANSI Std. B16.18. The solder used to make joints should be compatible with service conditions, and should not be used on pipe lines that carry flammable gases or liquids. Maximum working pressures and temperatures are according to table VII.

TABLE VII. Maximum working pressure and temperature, solder joint end valves

Solder	Valve class-	Temperature		Pressure rating, water tube size					
				1/4 to 1 inch		1-1/4 to 2 inches		2-1/2 and 3 inches	
		°F	°C	p.s.i.	Bars	p.s.i.	Bars	p.s.i.	Bars
Tin	A,B	100	37.8	200	13.8	175	12.1	150	10.6
lead	A,B	150	65.6	150	10.8	125	8.6	100	6.9
alloy	A,B	200	93.3	100	6.9	90	6.2	75	5.2
(50-50)	A,B	250	121.1	85	5.9	75	5.2	50	3.4
Tin	A	100	37.8	200	13.8	200	13.8	200	13.8
antimony	A	150	65.6	200	13.8	200	13.8	200	13.8
alloy	A	200	93.3	185	12.4	185	12.8	185	12.8
(95-5)	A	250	121.1	170	11.4	170	11.7	150	10.3
	B	100	37.8	300	20.7	300	20.7	300	20.7
	B	150	65.6	300	20.7	300	20.7	275	19.0
	B	200	93.3	270	17.2	250	17.2	200	13.8
	B	250	121.1	200	13.8	175	12.1	150	10.3

6.6 Civil agency procurements. When level B packaging is required for civil agency procurements, the level A in 5.1 shall apply.

6.7 Types I and II, stem packing. Valves are ordinarily shipped with the stem packing uncompressed. The packing nut should be tightened prior to placing valves in service.

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MILITARY INTERESTS:

Custodians:

Army - ME
Navy - YD
Air Force - 82

Review activity:

Army - ME, CE

User activity:

Navy - MC

Preparing activity:

GSA-FSS

Civil Agency Coordinating Activities:

GSA-PCD
HEW-FEC
INT-BPA
NASA-JFK
USDA-AFS

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